The Bodhi Seed

Twelve rhombic triacontahedra and one more inside a triacontahedron φ^2 times larger, with the gap filled exactly

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Version 1 · 12 October 2026 · Licence CC BY 4.0 · Data, scripts, renders and checks are included with this record and at https://github.com/DICTOR-Master/kaleidohedra.
The question, the golden-rhombohedron lead, the names Bodhi seed, Vajra, Crown Jewel and Diadem, the choice of builds and the hand-built twelve (Polyhedraverse) are DICTO's. Computation and checking with Claude (Anthropic).

Abstract. A rhombic triacontahedron RT(φ^2) of edge φ^2 holds twelve RT(1) of edge 1, one on each five-fold axis and each pushed out until it touches the outer form, and one RT(1/φ) at its centre, without overlap. Each of the twelve touches five others face to face, in the pattern of the icosahedron's 30 edges. These are the counts and ratios of the Lotus seed (an icosahedron of edge φ^2 holding 12 icosahedra of edge 1 and one of edge 1/φ), and among the icosahedral solids tested on their 5-, 3- and 2-fold axes only the icosahedron and the rhombic triacontahedron close in this way. The gap, 31.8% of the volume (by volume 80 prolate and 20 oblate golden rhombohedra of edge 1, though these do not tile it), fills exactly with convex pieces in two ways, each of three types and each invariant under the full icosahedral group: kit A, 60 + 60 + 60 mirror-symmetric pieces; kit B, 60 Vajras (triangular bipyramids), 20 Crown Jewels (one in each outer three-fold corner, 14 corners) and 60 + 60 Diadems of opposite hands. Every corner is exact and every dihedral angle is a multiple of 6°. The twelve enclose exactly a rhombic hexecontahedron of edge 1 (20 prolate golden rhombohedra), so the inner pocket, the hexecontahedron minus the centre, has exactly the volume of RT(1). All pieces were checked pairwise free of overlap, with total volume exact. The layer of twelve triacontahedra around a rhombic hexecontahedron appears in S. Kabai's clusters; its closure inside RT(φ^2), the centre, the exact kits and the growth as a seed were not found in the literature searched.

1. Which forms multiply

For each icosahedral form X (icosahedron, dodecahedron, icosidodecahedron, truncated icosahedron, rhombic triacontahedron) and each family of symmetry axes (12 five-fold, 20 three-fold, 30 two-fold directions), copies X(1) in the same turn were pushed out along every axis until they touched X(φ^k), k = 1 … 4, and kept when no two overlapped; then the largest centred copy was found. Only two cases close at golden sizes, with copies that fit and a centre that is exactly a power of φ: the icosahedron on its five-fold axes in I(φ^2) with centre I(1/φ) (the Lotus seed), and the rhombic triacontahedron on its five-fold axes in RT(φ^2) with centre RT(1/φ). Everything else either overlaps at φ and φ^2, or at φ^3 and φ^4 leaves a centre whose size is not a power of φ (`scripts/seeding.py`).

2. The twelve

The twelve copies' centres lie φ^2 from the middle, at the corners of an icosahedron of edge 2.752764 (twice the inradius of RT(1), 1.376382), so neighbours touch face to face: 30 contacts, five per copy. DICTO built the twelve by hand in the Polyhedraverse app; the

exported build has every centre at the computed distance to six decimals, 30 face contacts and no overlap. Volumes: RT(φ^2) 220.907161, twelve RT(1) 147.728810, RT(1/φ) 2.906170, gap 70.272181.

3. The hollow inside the twelve

The twelve enclose exactly a rhombic hexecontahedron RH(1) of edge 1: twenty prolate golden rhombohedra, one on each three-fold axis. The centre RT(1/φ) sits inside it, and the inner pocket RH(1) − RT(1/φ) has volume $20\varphi O - 10O/\varphi = 10\varphi^2 O$, exactly that of RT(1), where O is the oblate golden rhombohedron of edge 1 (volume 0.470228) and the prolate one is φO; in short RH(1) = RT(1) + RT(1/φ) by volume. The inner pocket is 60 Vajras; three Vajras fill each of the hexecontahedron's rhombohedra outside the centre, and that union is not convex, so no single three-fold inner piece exists.

4. Two kits

Piece	Kit	Count	Volume	Shape
Vajra	A, B	60	$\varphi^2 O/6$ = 0.205179	triangular bipyramid; edges 1, 1/φ, 1.0515, 1.4734, 1.7013; dihedrals 72°, 90°, 108°, 120°
Crown Jewel	B	20	$(7\varphi - 3)O/2$ = 1.957616	14 corners, 9 faces: three pentagons on RT(φ^2), three hexagons on the three copies around its corner, three cut triangles; dihedrals 36°, 90°, 120°, 144°
Diadem L, R	B	60 + 60	$O/3$ = 0.156743	6 corners, 5 faces, a rhombus face (edge 1) on a copy; chiral; dihedrals 36°, 60°, 90°, 108°
B2	A	60	$(3 + 5\varphi)O/6$ = 0.869152	7 corners: a pentagon on RT(φ^2), two quadrilaterals on copies, three cut triangles
B3	A	60	$O/(3\varphi)$ = 0.096872	pyramid on a golden rhombus (edge 1) lying on a copy; dihedrals 36°, 120°, 144°

Exact bookkeeping: gap = 80P + 20O; outer pocket = 70P + 10O, inner pocket = 10P + 10O = RT(1). Kit A uses 180 pieces, all mirror-symmetric; kit B 200 pieces, with one Crown Jewel in each three-fold corner as FUJI sits in the Lotus seed. Both kits were found by cutting one fundamental region (1/120) of the gap along the solids' face planes and the mirror planes, then choosing the fewest piece types and pieces by exact cover; three types are minimal for such cuts, and at least two are needed since the gap has two separate pockets.

5. Checks

For both kits: no overlapping pair among pieces and solids (separating axes); total volume equal to the gap to 6×10^{-14}; 236,269 random points each in exactly one solid or piece; all copies of a type congruent; the kit invariant under the 120 symmetries. Independently, for kit B: 235,529 sample points in RT(φ^2), each covered exactly once by the 13 solids and 200 pieces.

6. Growth

Since every triacontahedron inside is whole, each can be replaced by its own Bodhi seed, as the Chain of Karma does with Lotus seeds, the pieces staying whole: a Bodhi chain with 13^n triacontahedra after n steps. With the same counts and ratios as the Lotus seed, the triacontahedra alone form a self-similar set of the same dimension, D with $12\varphi^{-2D} + \varphi^{-3D} = 1$, D ≈ 2.6063.

7. Relation to known work

The rhombic triacontahedron is ten prolate and ten oblate golden rhombohedra, and the rhombic hexecontahedron twenty prolate ones (classical). S. Kabai built clusters of rhombic triacontahedra and hexecontahedra as models of quasicrystal structure, among them a layer of twelve triacontahedra face to face around a hexecontahedron (Wolfram Demonstrations Project, "Cluster of 45 Polyhedra around a Rhombic Hexacontahedron" and "Dodecahedral Cluster of RTs and RHs"; Kabai, Bérczi and Szilassi). We found no description of that layer closing exactly inside RT(φ^2) with RT($1/\varphi$) at its centre, of a convex exact filling of the gap, or of its growth as a seed. The search is not exhaustive.

Figures

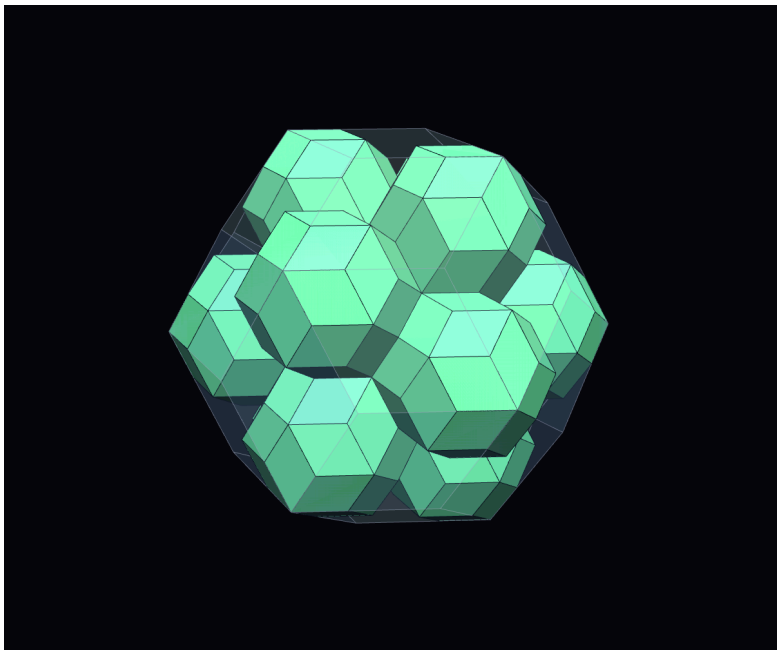


Figure 1. The Bodhi seed: RT(φ^2) (crystal) holding twelve RT(1) (green) around RT($1/\varphi$) (gold).

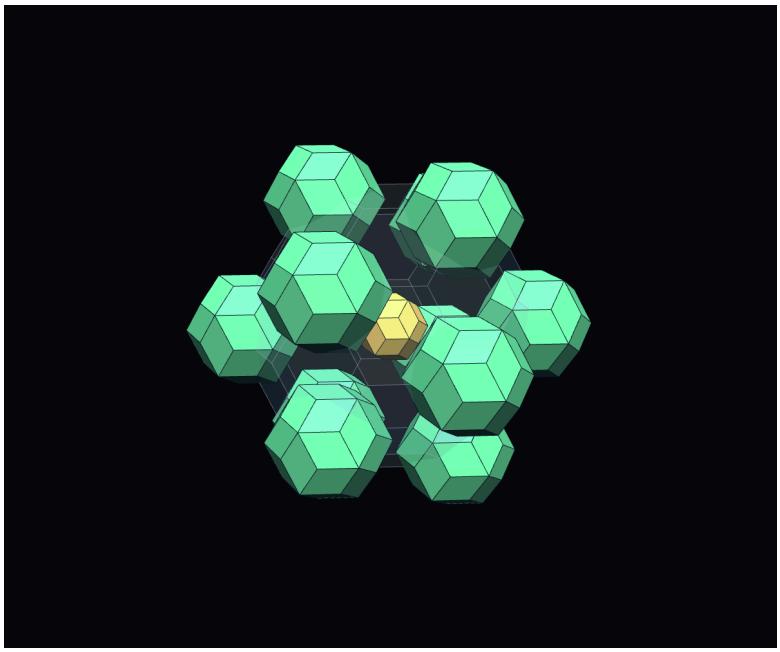


Figure 2. The thirteen pulled apart.

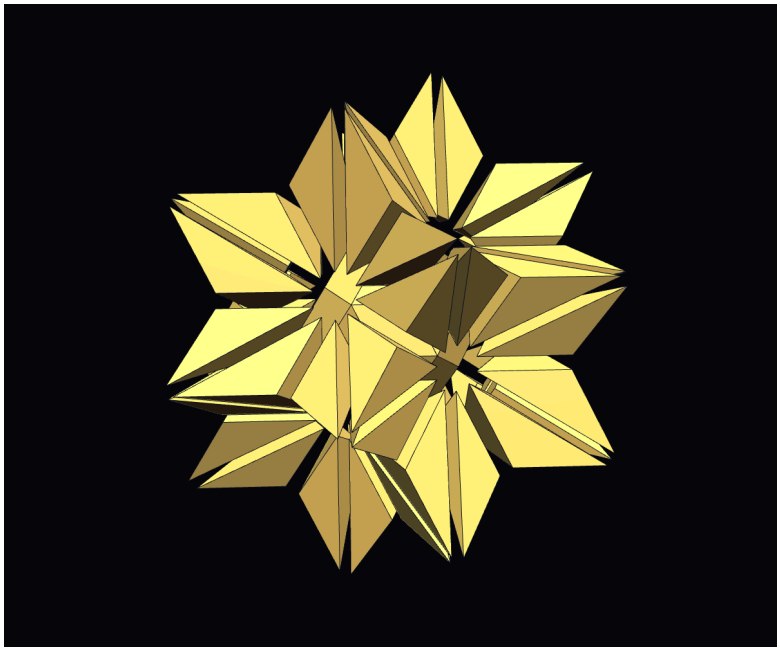


Figure 3. 60 Vajras around the centre: the rhombic hexecontahedron, slightly pulled apart.

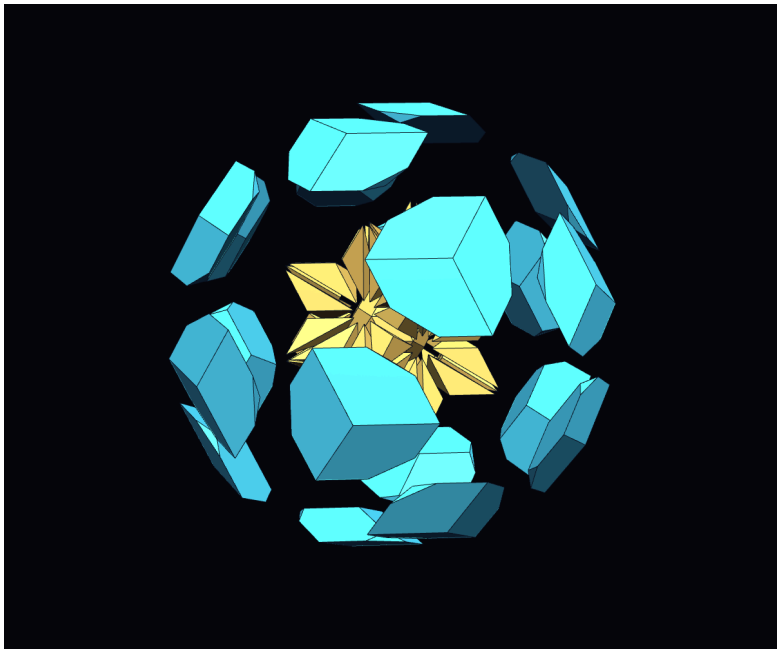


Figure 4. The 20 Crown Jewels (blue) around the Vajras.

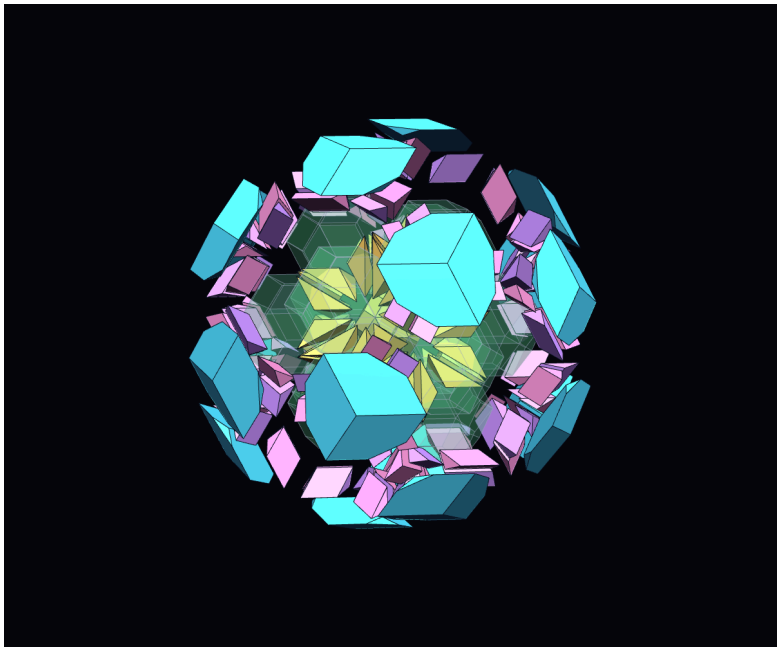


Figure 5. Kit B pulled apart: Vajras (gold), Crown Jewels (blue), Diadem L (lilac), Diadem R (pink); the twelve faint.

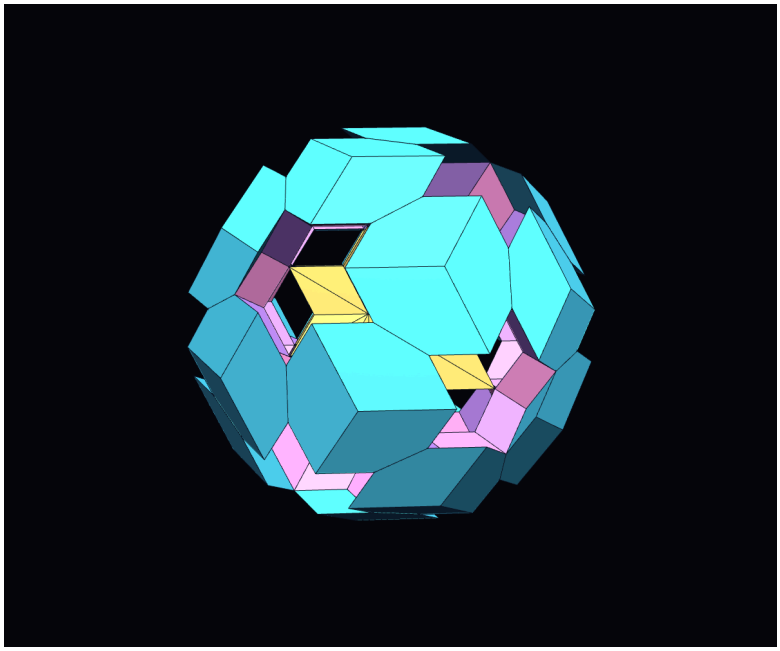


Figure 6. Kit B closed, without the twelve: the gap, with its twelve sockets.

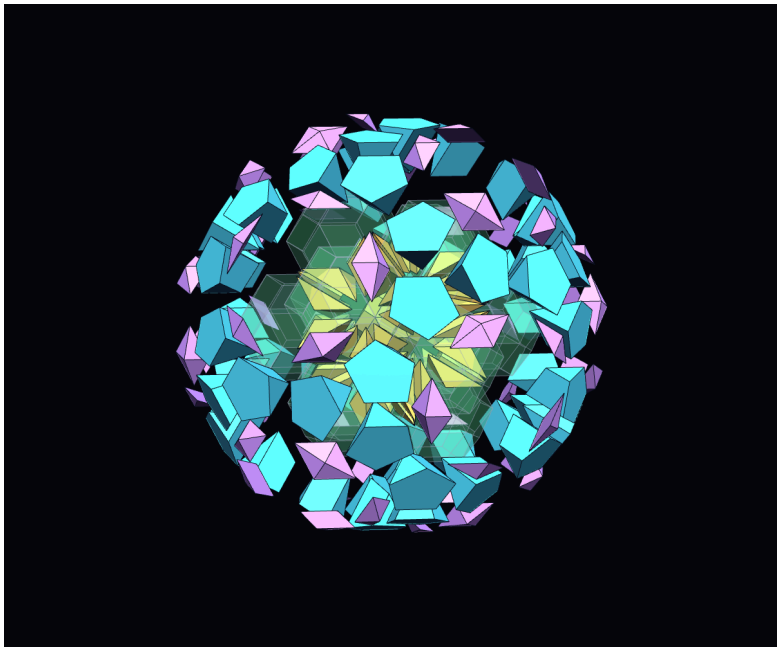


Figure 7. Kit A pulled apart: Vajras, B2 (blue), B3 (lilac).

References

1. S. Kabai, Cluster of 45 Polyhedra around a Rhombic Hexacontahedron; Dodecahedral Cluster of RTs and RHs. Wolfram Demonstrations Project.
2. S. Kabai, Sz. Bérczi, L. Szilassi, on clusters of rhombic triacontahedra and hexecontahedra, The Mathematica Journal.
3. DICTO, The Chain of Karma (2026), doi:10.5281/zenodo.23310542.
4. DICTO, Spheres of Dharma (2026), doi:10.5281/zenodo.23310834.